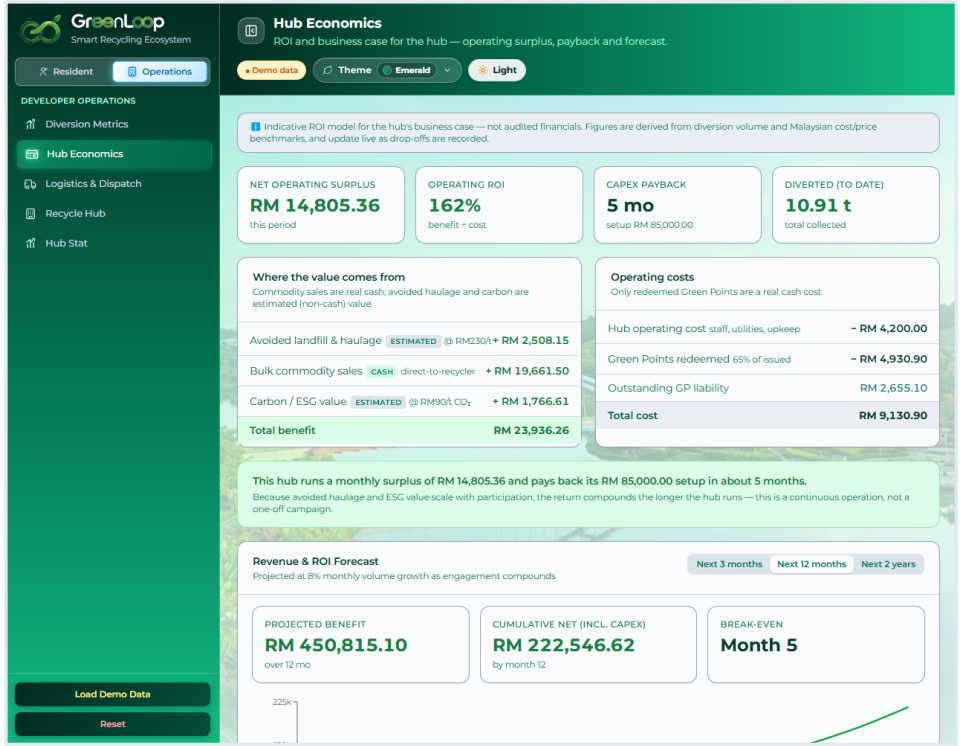
Appendix

GreenLoop · Team Synkosys · supporting detail for the pitch deck

A Business case & assumptions

Indicative Malaysian benchmarks (2024). Stated openly, not audited financials.

Avoided landfill & haulage	**RM 230 / tonne diverted**
Carbon / ESG value	**RM 90 / tonne CO_2e**
Hub operating cost (opex)	**RM 4,200 / month**
One-time setup (CapEx)	**RM 85,000**
Green Point value	**RM 0.10 per GP (10 GP = RM 1)**
GP redemption rate	**65% of issued GP redeemed for cash**
Assumed volume growth	**8% per month**
Bulk price: aluminium / e-waste	**RM 4.2 / RM 12 per kg**
Bulk price: PET / paper / cardboard	**RM 1.1 / 0.55 / 0.45 per kg**



B AI & system status · honest boundary

What is genuinely working today vs. what still needs building.

WORKING TODAY

✓ Working in the browser

- TensorFlow.js COCO-SSD on webcam / uploaded photos
- Green Points wallet, redemption & fee offset
- Dispatch sim on real roads (OSRM) + 6-hub map



~ Simulated in the prototype

- Bin fill levels & sensor weights (demo values)
- Two-camera cheat check (entry vs inside)
- Township ROI from an indicative, tunable model



WHAT STILL NEEDS BUILDING

~ Mobile Application

- Phone Apps for User
- User to connect with the Smart Recycle Hub
- Activity monitoring + Green Points wallet, redemption & fee offset

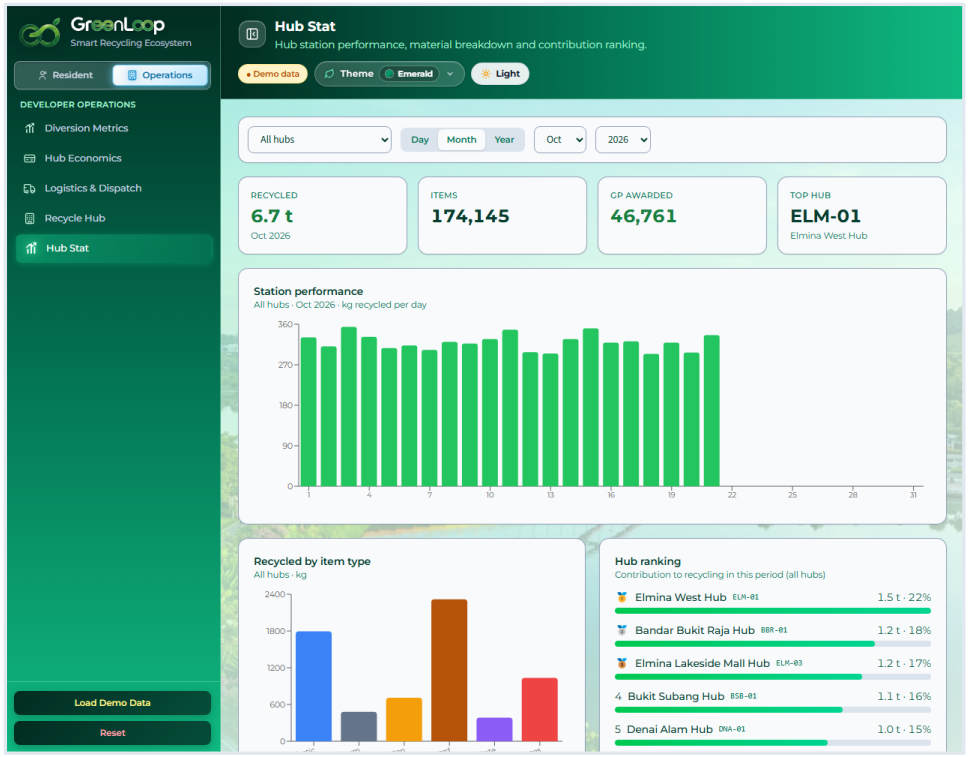


! Needs real-world build

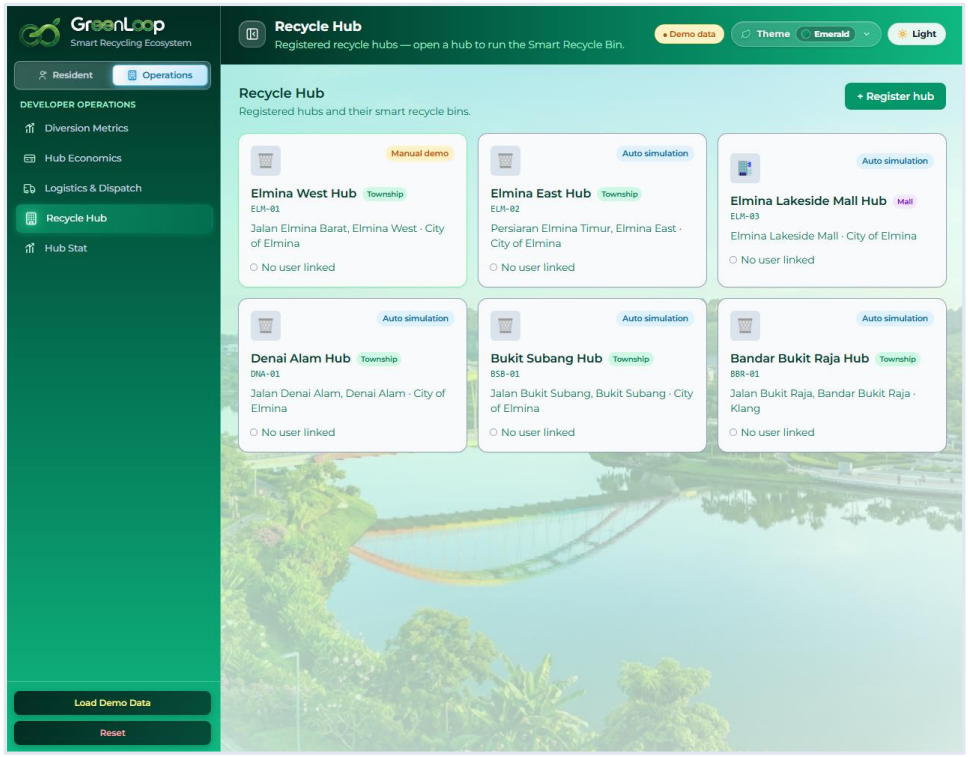
- Recyclability/material classifier on real items
- Physical smart-bin hardware + load cells
- POS/voucher + identity + telemetry integration

On AI accuracy & anti-cheat: the demo detects common objects (bottle, can, cup, phone); it does not yet confirm material purity. Production needs a recyclability classifier validated on real items, plus the two-camera entry/inside check to stop re-scanning without depositing.

C Operator analytics & network



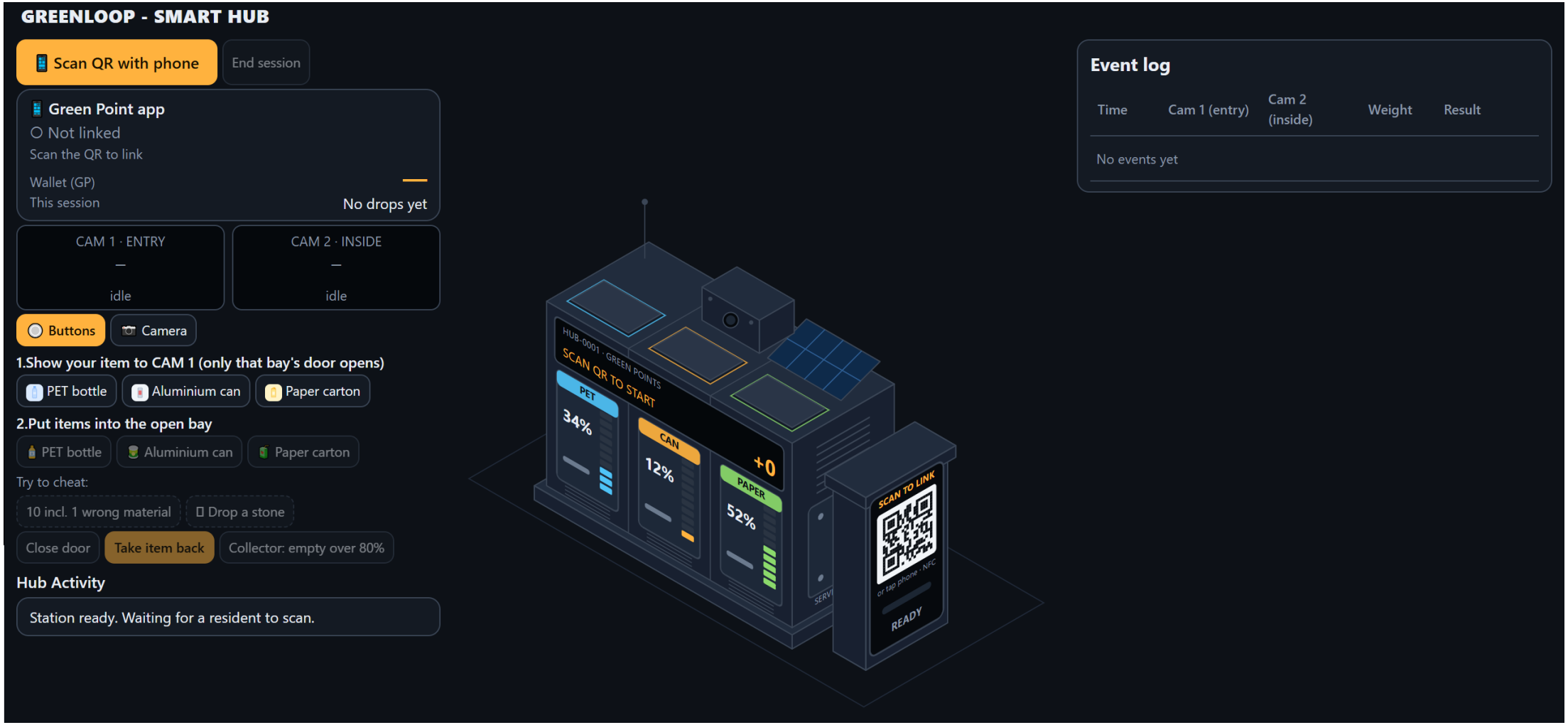
Hub Stat · day / month / year, respects 09:00 to 17:00 hours



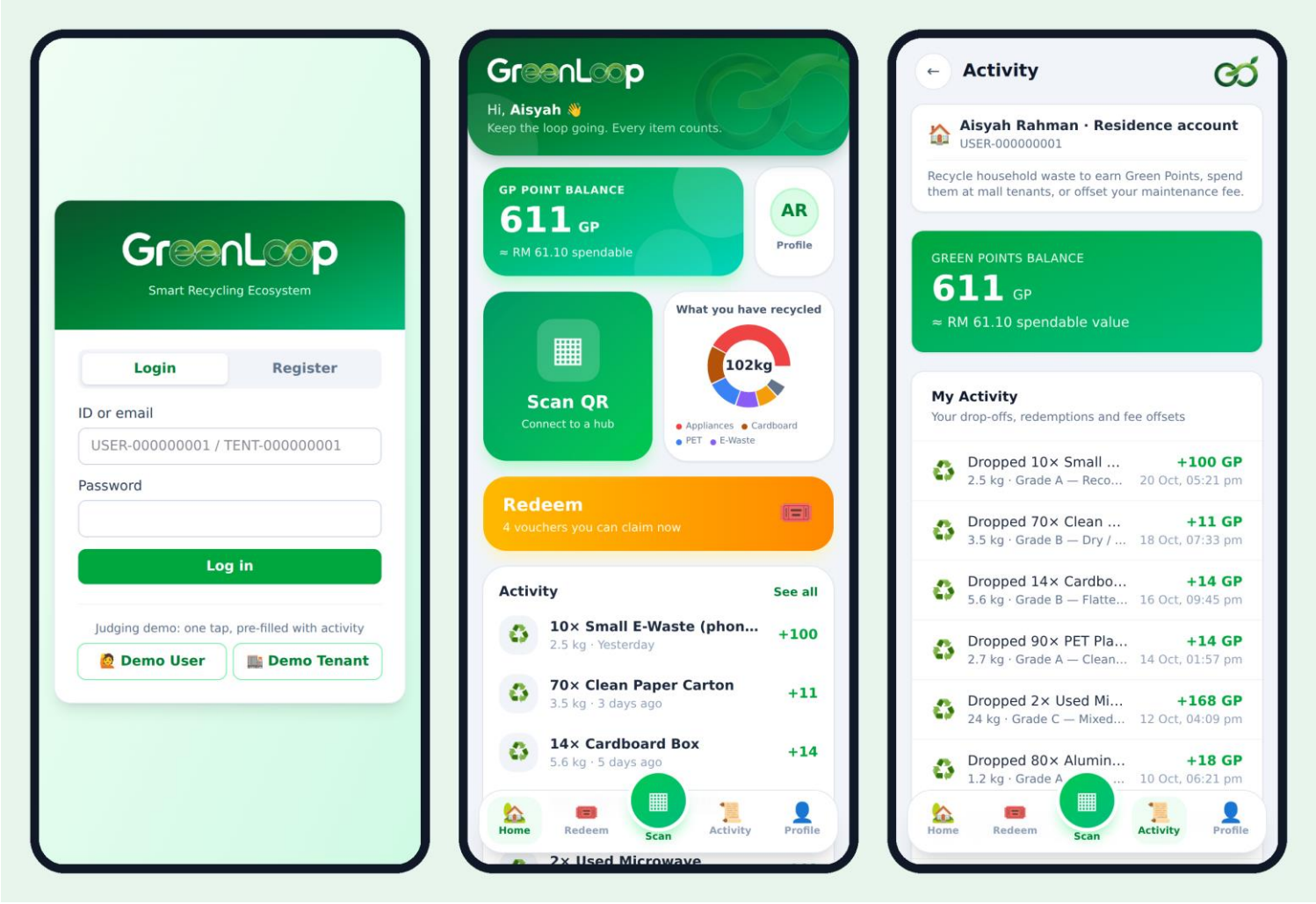
Recycle Hub network (6 hubs, City of Elmina + Bandar Bukit Raja)

Tech stack: React + Vite · Tailwind · TensorFlow.js COCO-SSD · OSRM road routing · Leaflet maps · Recharts. All client-side; no backend.

D Hub Simulation (Prototype Simulator)



E Mobile Apps (Proposal)



F How we meet the judging criteria

Impact	Environmental (diversion, CO_2), community (rewards) and operational (cost, visibility) outcomes.
Innovation	AI item verification + funded reward model + live AI dispatch in one loop.
Feasibility	Deployed working prototype; realistic one-hub pilot; clear validation checklist.
Strategic fit	Built on real Sime Darby Property townships; supports ESG and SHIFT25 goals.
Commercial value	Self-funding business case: operating surplus + CapEx payback, not a subsidy.
Team & pitch	A focused team shipping a real, honest, demoable product.